

ARTICLE PRESENTATION • GROUP 2

Trading Value and Information in MDPs

Jonathan Rubin, Ohad Shamir & Naftali Tishby (2012)

Presented by Yi Yao • Mingze • Jeethu

April 2026

What we'll cover today

01 Research Question

Why combine information with reward in MDPs?

02 Methodology

The mathematical framework — free energy & Bellman

03 Approach

The INFO-RL algorithm

04 Results

The value–information tradeoff curve

05 Findings

Stochastic policies, robustness & implications

Standard MDPs ignore the cost of control

The classical setup

- An agent interacts with its environment modeled as a Markov Decision Process
- Traditional goal: find a policy π that maximizes expected accumulated reward (value)
- Optimal policy is deterministic — act greedily with respect to $V^*(s)$

What's missing

- The circular flow of information between agent and environment is ignored
- Sending control signals through a channel (brain→muscles, controller→robot) is not free
- Deterministic policies are expensive in the bits needed to specify them

The central question

" *How can we formally combine the cost of control information with the reward measure — and what does the resulting tradeoff between value and information look like?*

Sub-questions addressed in the paper

- Can we derive a Bellman-style recursion that handles both value and information jointly?
- What shape does the tradeoff take, and does it justify stochastic policies?
- Does the information term double as a regularizer when reward estimates are noisy?

SECTION 02

Methodology

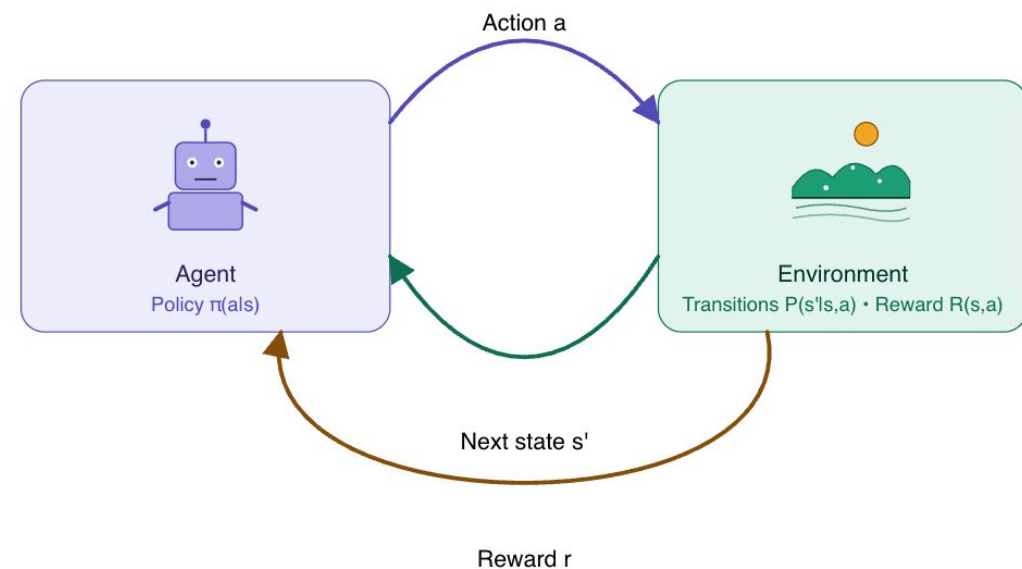
The mathematical framework — presented by Yi Yao

Main Ideas

- Standard MDP = maximize sequential decision making reward
- Decision making in information theory framework:
Communication Cost
- Tradeoff between **expected reward (value)** and **control information**
- More: regularization

Standard MDP Setup

- states
- actions
- rewards
- transition probabilities
- a stationary policy



Goal — stochastic shortest path: reach the terminal state with maximum expected reward
Each step costs $R < 0$ • terminal state is free

Goal

- narrow the setting to the **stochastic shortest path**

Setup (2)

- Value function of policy π
$$V_{\pi}(s_0) = \lim_{T \rightarrow \infty} \mathbb{E} \left[\sum_{t=0}^{T-1} R(s_t, a_t) \right]$$
- the probability of a whole trajectory under policy π

$$\Pr(a_0, s_1, a_1, s_2, \dots | s_0) = \prod_{t \geq 0} \pi_{s_t}(a_t) P_{s_t, a_t}(s_{t+1})$$

Setup (3)

- Optimal Value function
$$V^*(s) = \max_{\pi_s(\cdot)} \sum_a \pi_s(a) \sum_{s'} P_{s,a}(s') [R(s,a) + V^*(s')]$$
$$= \max_a \left[R(s,a) + \sum_{s' \in \mathcal{S}} P_{s,a}(s') V^*(s') \right]$$
- Optimal states-actions value function & optimal policy

$$Q^*(s,a) = R(s,a) + \sum_{s' \in \mathcal{S}} P_{s,a}(s') V^*(s')$$

$$\pi_s^*(a) = \begin{cases} 1 & a = \arg \max_{a'} Q^*(s,a') \\ 0 & \text{otherwise} \end{cases}$$

Next, introduce Control Information

- one-step control information cost at state s (KL divergence)

$$\Delta I(s) = \sum_a \pi_s(a) \log \frac{\pi_s(a)}{\rho_s(a)}$$

- assume $\rho_s(a)$ is uniform over actions

- total control information

$$\begin{aligned} I_\pi(s_0) &= \lim_{T \rightarrow \infty} \mathbb{E} [\Delta I(s_t)] \\ &= \lim_{T \rightarrow \infty} \mathbb{E} \left[\sum_{t=0}^{T-1} \log \frac{\pi_{s_t}(a_t)}{\rho_{s_t}(a_t)} \right] \\ &= \lim_{T \rightarrow \infty} \mathbb{E} \left[\log \frac{\Pr(a_0, a_1, \dots, a_{T-1} | s_0; \pi)}{\Pr(a_0, a_1, \dots, a_{T-1} | s_0; \rho)} \right] \end{aligned}$$

Free Energy Formulation

- free energy formulation function

$$\begin{aligned} F_{\pi}(s_0; \beta) &= I_{\pi}(s_0) - \beta V_{\pi}(s_0) \\ &= \lim_{T \rightarrow \infty} \mathbb{E} \sum_{t=0}^{T-1} \left[\log \frac{\pi_{s_t}(a_t)}{\rho_{s_t}(a_t)} - \beta R(s_t, a_t) \right] \end{aligned}$$

- minimization $\rightarrow$ constrained optimization problem

$$\max_{\pi} V_{\pi}(s_0) \quad s.t. \quad I_{\pi}(s_0) \leq threshold$$

Recall: Rate-distortion Theory

- Trade-off for:

Rate: The average number of bits per symbol used to describe the source information.

Distortion: A measure of the discrepancy between the original source and its reconstructed representation.

- Convexity
- Stochastic Nature

→ **Control Information as Bit Rate & Optimal Planning**

Bellman's optimality equation

$$[BF](s) = \min_{\pi_s(\cdot)} \sum_a \pi_s(a) \left[\log \frac{\pi_s(a)}{\rho_s(a)} - \beta R(s, a) + \sum_{s'} P_{s,a}(s') F(s'; \beta) \right]$$

- Unique fixed point $\rightarrow$ closed-form solution

$$[BF](s) = -\log Z(s; \beta)$$

$$Z(s; \beta) = \sum_a \rho_s(a) \exp [\beta R(s, a) - \mathbb{E}_{s'|s,a} F(s'; \beta)]$$

Closed-form solution

$$\pi_s(a) = \frac{\rho_s(a)}{Z(s; \beta)} \exp [\beta R(s, a) - \mathbb{E}_{s'|s, a} F(s'; \beta)]$$

$$Z(s; \beta) = \sum_a \rho_s(a) \exp [\beta R(s, a) - \mathbb{E}_{s'|s, a} F(s'; \beta)]$$

Iterative method with update

$$F_{k+1}(s; \beta) = [BF_k](s)$$

The INFO-RL algorithm

Dynamic programming on the modified Bellman equation

Input: MDP (S, A, P, R) , default ρ , tradeoff β

1. Initialize $F(s) \leftarrow 0$ for all $s \in S$

2. Repeat until F converges:

for each state s :

$Z(s; \beta) \leftarrow \sum_a \rho(a) \cdot \exp[\beta R(s, a) - E F(s')]$

$F(s; \beta) \leftarrow -\log Z(s; \beta)$

3. Extract optimal policy:

$\pi^*(a|s) \propto \rho(a) \cdot \exp[\beta R(s, a) - E F(s')]$

Return π^*

Key properties

- Guaranteed convergence to a unique global fixed point F^*
- Solutions are intrinsically stochastic (softmax form)
- β controls the tradeoff: $\beta \rightarrow \infty$ recovers greedy optimal; $\beta \rightarrow 0$ gives uniform random walk
- For deterministic transitions, reduces to a linear system $Z = \Phi Z$

Experimental setup: 10×10 grid world

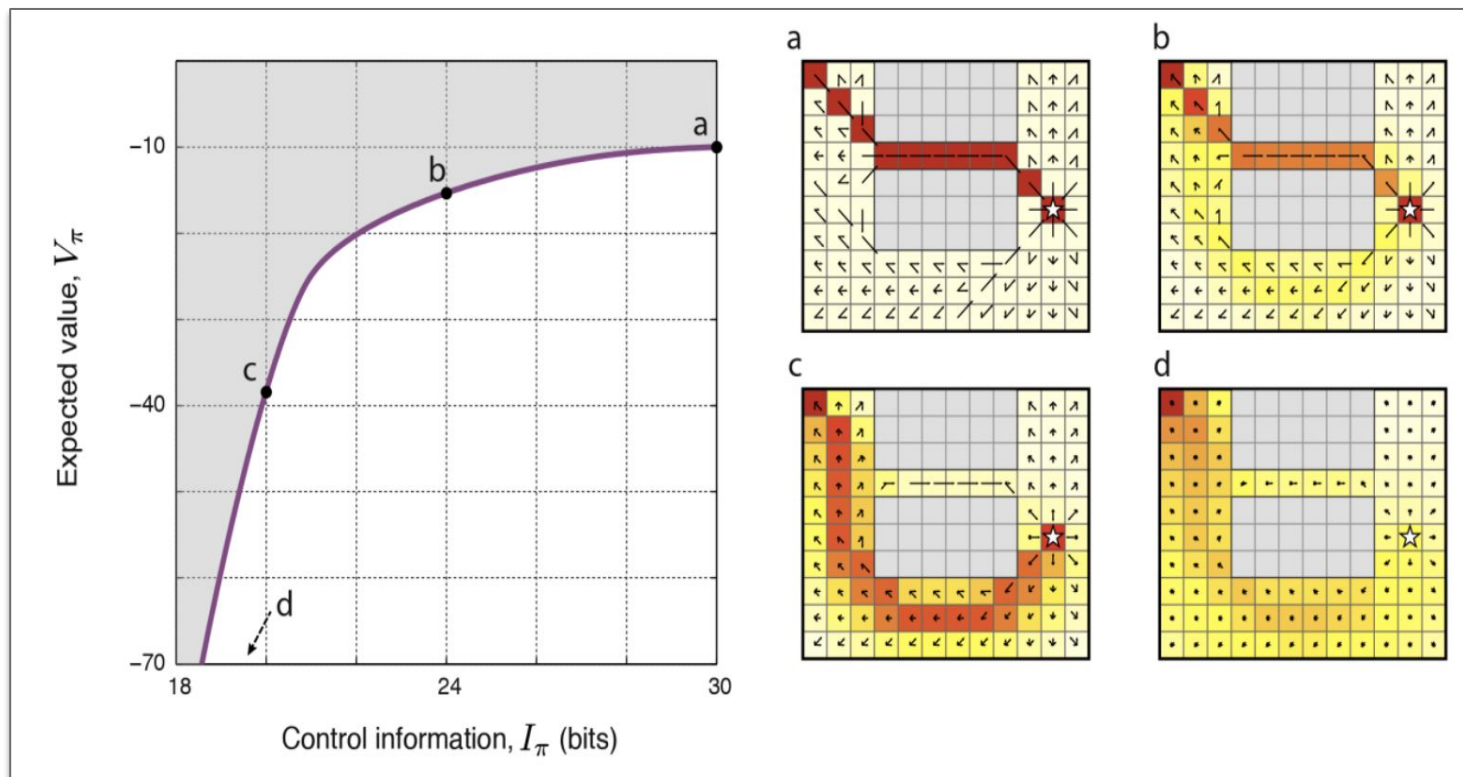
Navigation task

- Agent starts top-left, must reach a goal state while avoiding walls
- Eight possible actions per step: $\{\uparrow, \nearrow, \rightarrow, \searrow, \downarrow, \swarrow, \leftarrow, \nwarrow\}$
- Step cost: $R = -1$ • Wall collision: $R = -100$
- Deterministic transitions — the linear form of INFO-RL applies

Information accounting

- Each fully deterministic action costs $\log_2(8) = 3$ bits of control information
- A 10-step deterministic path to the goal therefore costs $10 \times 3 = 30$ bits
- INFO-RL is run across many values of β to trace out the full value–information tradeoff

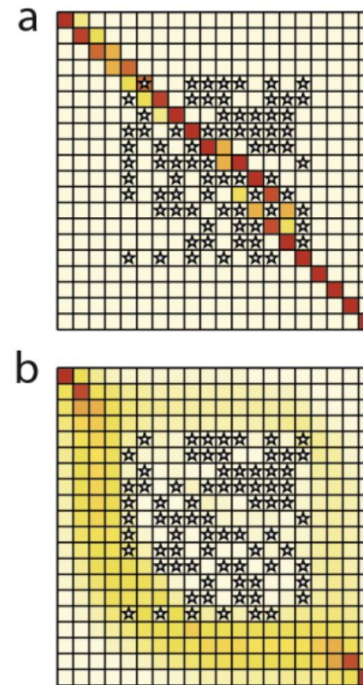
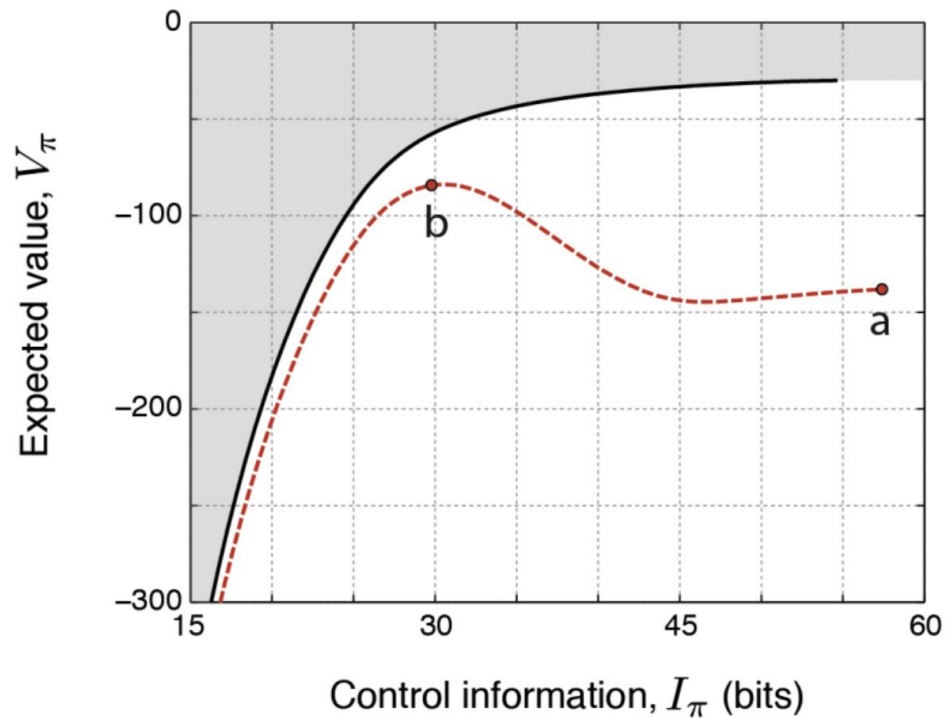
The value–information tradeoff curve



Key observations

- The curve is concave — analogous to rate-distortion
- Above the curve: infeasible region (no such policy exists)
- Below the curve: suboptimal policies
- The relationship is far from linear — policies can give up very little value while paying much less information

Robustness under noisy rewards



Setup

- Mines punish $r = -20$ with 50% probability
- Agent only sees one noisy sample of the reward function
- $\beta \rightarrow \infty$ policy overfits — tries to squeeze between sampled mines

Finding

- Lower β regularizes — policy routes around the field entirely
- The info term acts as principled regularization against overfitting

Main takeaways

- ▶ **A principled Bellman recursion for information**

The paper derives the first Bellman-type recursive equation unifying value and control information through a free-energy objective.

- ▶ **Justification for stochastic policies**

Unlike softmax action selection — bolted on after solving for Q^* — here stochastic policies emerge naturally from the optimization itself.

- ▶ **A concave value–information tradeoff**

Small sacrifices in expected reward yield dramatic savings in control information — useful for bandwidth-limited and bounded-rational agents.

- ▶ **Information as a regularizer**

Via a PAC-Bayes bound, the control information term is a principled regularizer that improves generalization under noisy rewards.

Why this paper still matters

Impact on modern reinforcement learning

- Foundational for the control-as-inference and KL-regularized RL literature
- Direct conceptual predecessor of entropy-regularized / maximum-entropy RL (e.g. Soft Actor-Critic)
- Connects RL to rate-distortion theory and the information bottleneck framework
- Relevant to bounded-rational decision making, neuroscience, and bandwidth-limited robotics

Limitations noted by the authors

- Focuses only on the control (action) information — the perception term is left for future work
- Requires a known MDP model; not directly a model-free learning algorithm

Discussion

1. The paper justifies control information as the literal cost of sending commands through a channel — brain to muscles, controller to robot. Is this a compelling physical motivation, or is it really just entropy regularization dressed up in an information-theory story? Would the framework lose anything if we dropped the channel metaphor entirely?
2. The modified Bellman operator has a unique fixed point, and in the deterministic-transitions case the whole thing collapses into a linear system $Z = \Phi Z$. Why does stochasticity in the transitions reintroduce nonlinearity, and what does that tell us about where the actual computational difficulty of the problem lives?
3. The same control information term that represents a "channel cost" in Section 3 suddenly reappears in Section 5 as a KL regularizer against overfitting noisy rewards. Do you read this as a deep unification — one object playing two roles — or as a mathematical coincidence that got a nice narrative wrapped around it? And does the regularization argument still matter in modern RL where agents collect millions of samples rather than one?
4. This paper is a conceptual ancestor of Soft Actor-Critic and the whole maximum-entropy RL line that dominates deep RL today. What did those later methods keep from Rubin et al., and what did they quietly drop? And if you were updating the framework for 2026, would you be more excited to tackle the perception half of the information loop (which the authors explicitly left open), or to make it work without a known MDP model?

Thank you!

Questions & discussion

Rubin, J., Shamir, O., & Tishby, N. (2012). Trading Value and Information in MDPs.

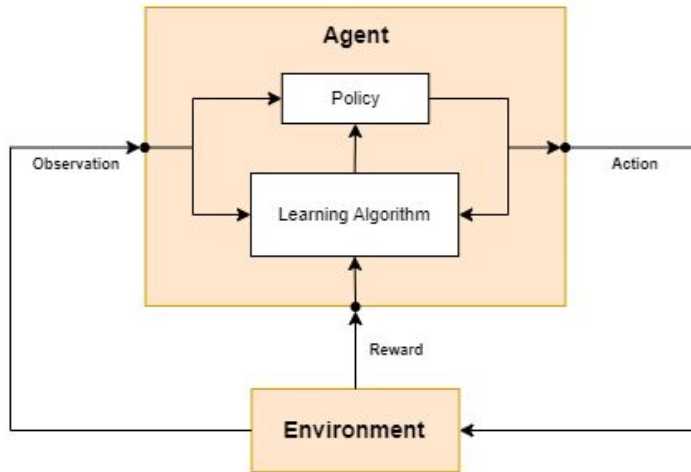
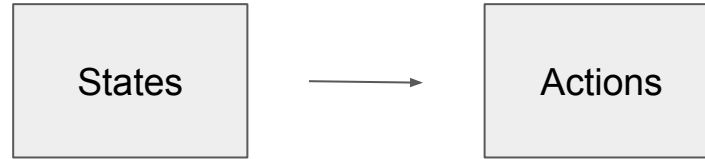
In Decision Making with Imperfect Decision Makers, Springer.

RESEARCH ARTICLE

Human decision making balances reward maximization and policy compression

Lucy Lai ^{1,2*}, Samuel J. Gershman³

Decision making is **mapping** states to actions



Reinforcement learning:
 $\pi(a|s)$ as mapping strategy - “**policy**”

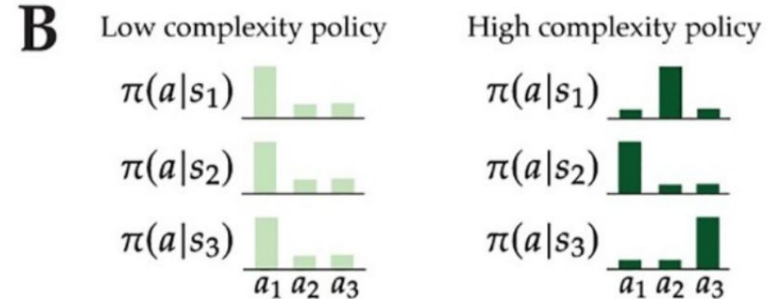
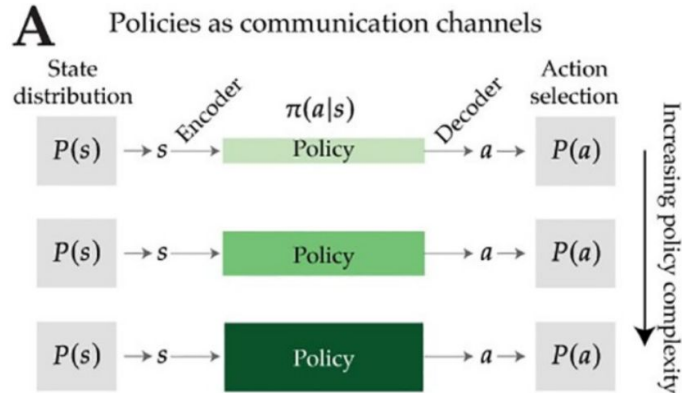
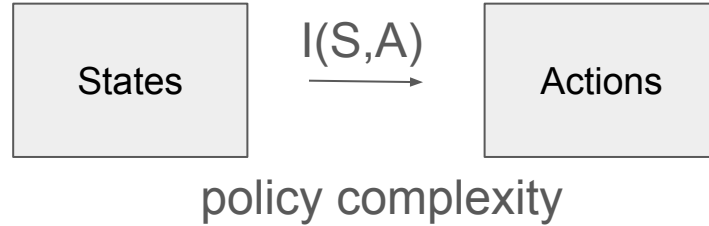
Deterministic:

$$\pi^* = \operatorname{argmax} Q(s,a)$$

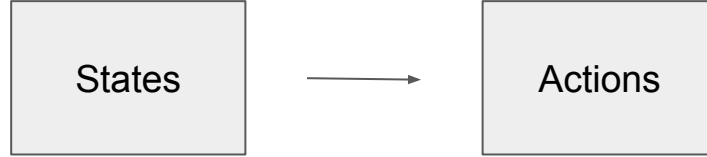
Stochastic: standard softmax

$$\pi^* \propto \exp[\beta Q(s,a)]$$

Decision making in **information theory** framework

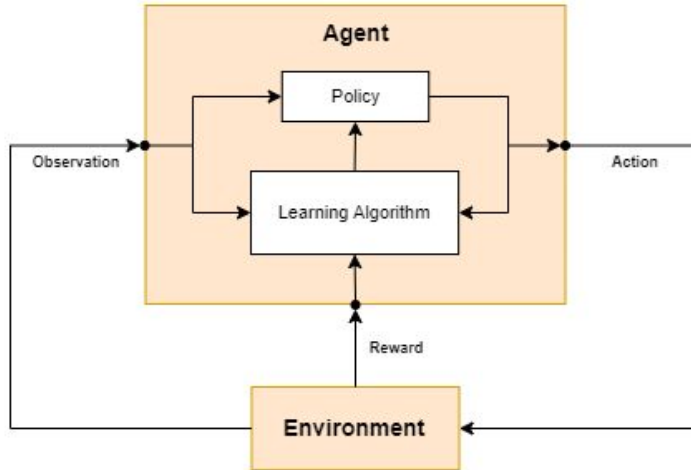
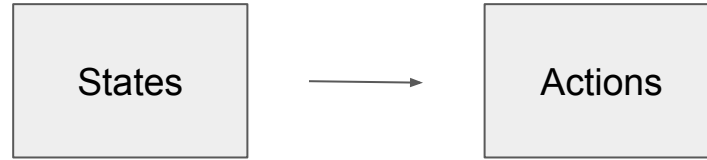


Biological brains have **limited** cognitive resources



finite cognitive resource

Decision making is **mapping** states to actions



Reinforcement learning:
 $\pi(a|s)$ as mapping strategy - “**policy**”

Deterministic:

$$\pi^* = \operatorname{argmax} Q(s,a)$$

Stochastic: standard softmax

$$\pi^* \propto \exp[\beta Q(s,a)]$$

Decision making in **information theory** framework: for **biological** brain

Standard RL: In **each** state, maximize reward while staying somewhat **random**

Standard softmax:

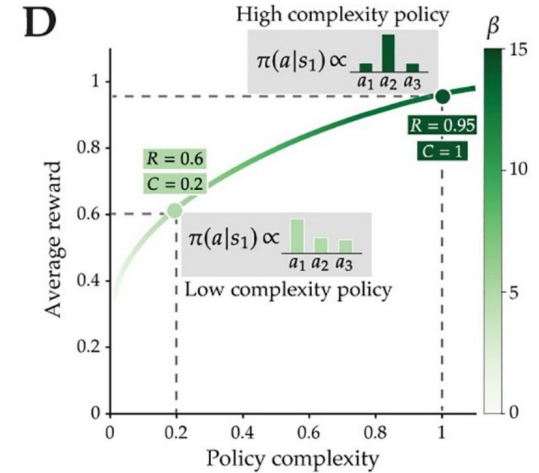
$$\pi^* \propto \exp[\beta Q(s,a)]$$



Policy compression: Across **all** states jointly, maximize reward while keeping the action distributions across states **similar to each other**

$$I^\pi(S; A) = \sum_s P(s) \sum_a \pi(a|s) \log \frac{\pi(a|s)}{P(a)},$$

$$\pi^*(a|s) \propto \exp[\beta Q(s, a) + \log P^*(a)],$$



(9)

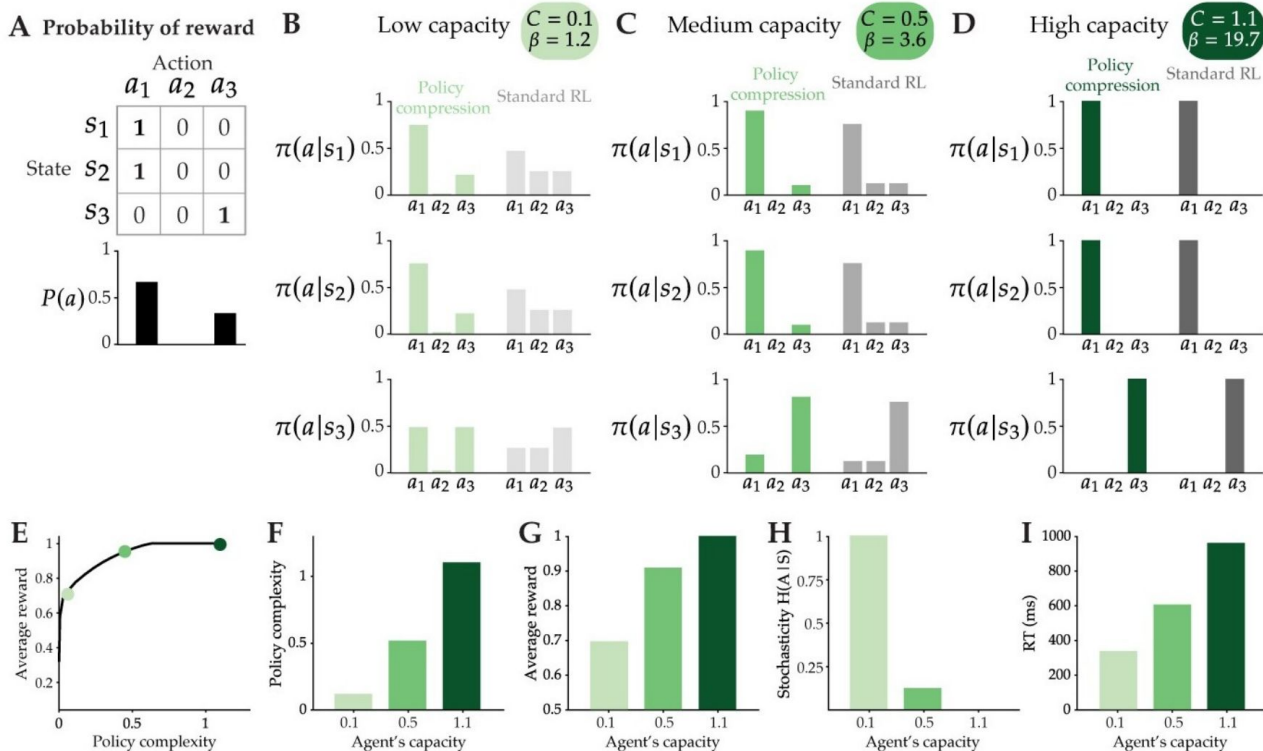
C

$$\pi(a|s_1) \propto \exp \left[\beta \frac{Q(s, a)}{a_1 \ a_2 \ a_3} + \log \frac{P(a)}{a_1 \ a_2 \ a_3} \right]$$

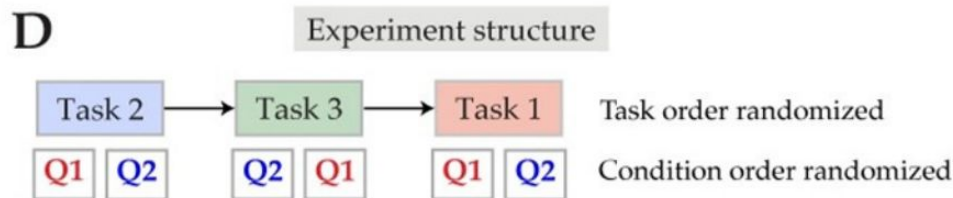
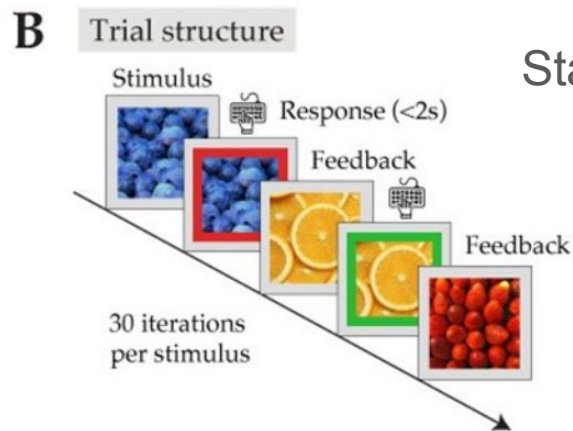
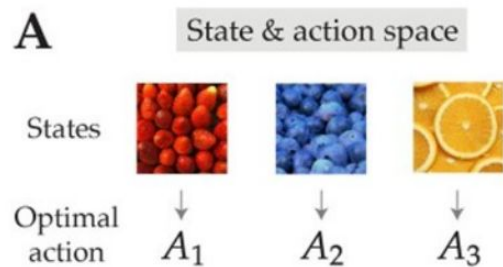
$$\pi^* \propto \exp[\beta Q(s,a)]$$

$$\pi^*(a|s) \propto \exp[\beta Q(s,a) + \log P^*(a)]$$

Experiment: Standard RL vs. Policy Compression Models



Experimental Design - Humans



Model Fitting

Table 1. List of models and their free parameters. Models vary in whether they penalize policy complexity and how they update β .

No.	Model	β update rule	Parameters
1	RLWM	n/a	$C, \alpha_{RL}, \phi, \rho, \gamma, A, B_1, B_2, \eta$
2	Standard RL: 1β	n/a	$\beta, \alpha, A, B_1, B_2, \eta$
3	Standard RL: 6β	n/a	$\beta_{11}, \beta_{12}, \beta_{21}, \beta_{22}, \beta_{31}, \beta_{32}, \alpha, A, B_1, B_2, \eta$
4	Fixed: 1β	n/a	$\beta, \alpha_\theta, \alpha_V, \alpha_P, b_1, b_2$
5	Fixed: 6β	n/a	$\beta_{11}, \beta_{12}, \beta_{21}, \beta_{22}, \beta_{31}, \beta_{32}, \alpha_\theta, \alpha_V, \alpha_P, b_1, b_2$
6	Adaptive: Capacity	$\Delta\beta = \alpha_\beta[C - \xi]$	$C, C_{\text{reduced}}, \beta_0, \alpha_\beta, \alpha_\theta, \alpha_V, \alpha_P, b_1, b_2$
7	Adaptive: Value	$\Delta\beta = \alpha_\beta[R - \rho]$	$R, R_{\text{reduced}}, \beta_0, \alpha_\beta, \alpha_\theta, \alpha_V, \alpha_P, b_1, b_2$
8	Adaptive: Capacity-Value	$\Delta\beta = \alpha_\beta \left[\frac{C-\xi}{R-\rho} - \beta \right]$	$C, C_{\text{reduced}}, R, \beta_0, \alpha_\beta, \alpha_\theta, \alpha_V, \alpha_P, b_1, b_2$

<https://doi.org/10.1371/journal.pcbi.1012057.t001>

Table 2. Parameter bounds for the RLWM and standard RL models.

Parameter	RLWM	Standard RL: 1β	Standard RL: 6β
C	[2, 5]	-	-
β	-	[1, 30]	all 6β s $\in$ [1, 30]
α	[0, 1]	[0, 1]	[0, 1]
ϕ	[0, 1]	-	-
ρ	[0, 1]	-	-
γ	[0, 1]	-	-
A	[0, 500]	[0, 500]	[0, 500]
$B_1 - A$	[0, 500]	[0, 500]	[0, 500]
$B_2 - A$	[0, 500]	[0, 500]	[0, 500]
η	[0, 3]	[0, 3]	[0, 3]
Total # Parameters	9	6	11

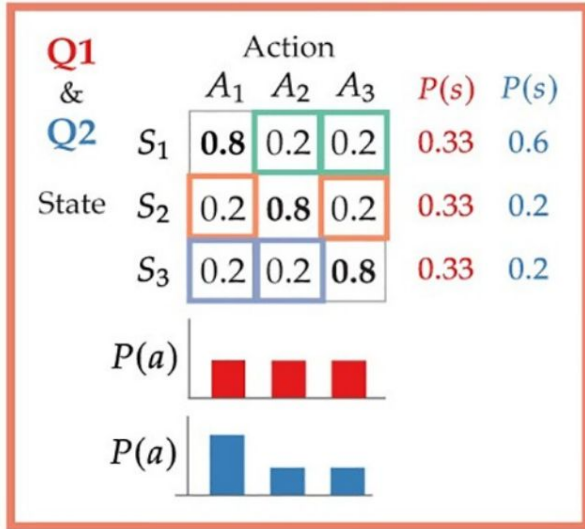
<https://doi.org/10.1371/journal.pcbi.1012057.t002>

Across all three tasks, the “Adaptive: Capacity-Value” model came closest to capturing the dynamics of learning in subjects’ data in all three tasks. Though this model scored close to the “Adaptive: Capacity” and “Adaptive: Value” models in quantitative model comparison metrics, the other two adaptive models could not capture key aspects of the learning dynamics in Tasks 2 and 3 (for a more detailed explanation, please refer to [S3](#) and [S4](#) Figs, and their captions).

Does State Frequency Bias Actions?

A

Task 1



Q1 control, Q2 experimental (S^* appears 3X more often)

Does State Frequency Bias Actions?

A

Task 1

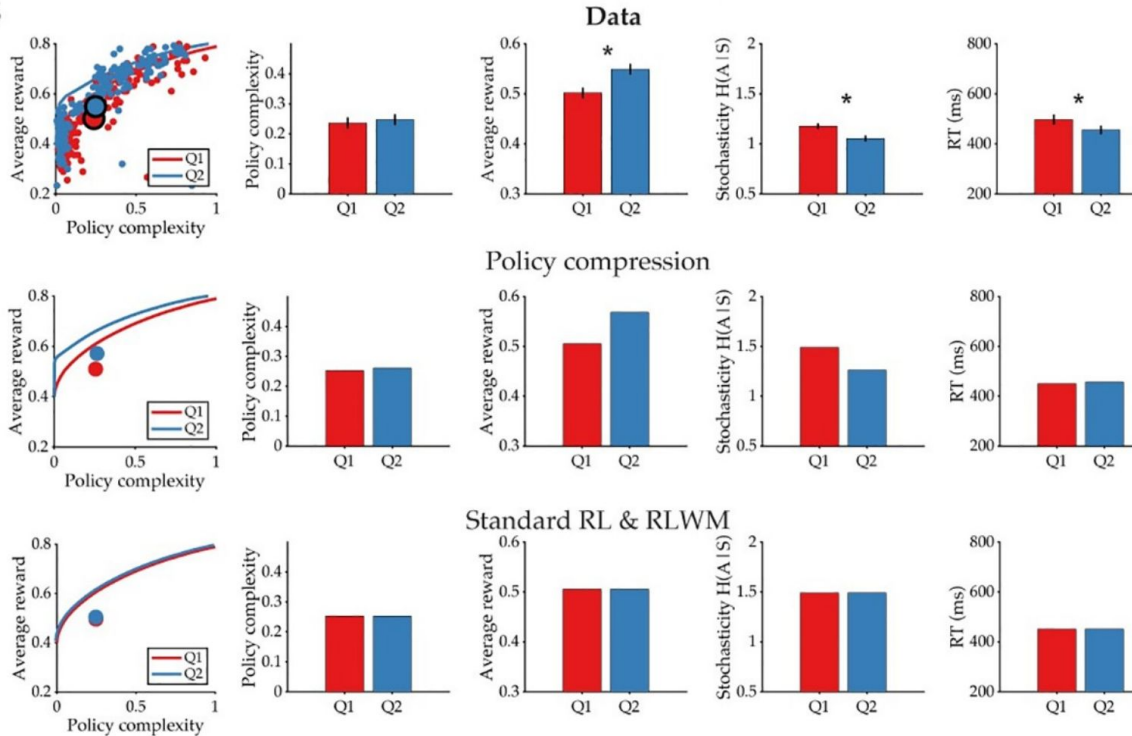
		Action			$P(a)$	$P(s)$
		A_1	A_2	A_3		
State	S_1	0.8	0.2	0.2	0.33	0.6
	S_2	0.2	0.8	0.2	0.33	0.2
	S_3	0.2	0.2	0.8	0.33	0.2

$P(a)$

$P(s)$

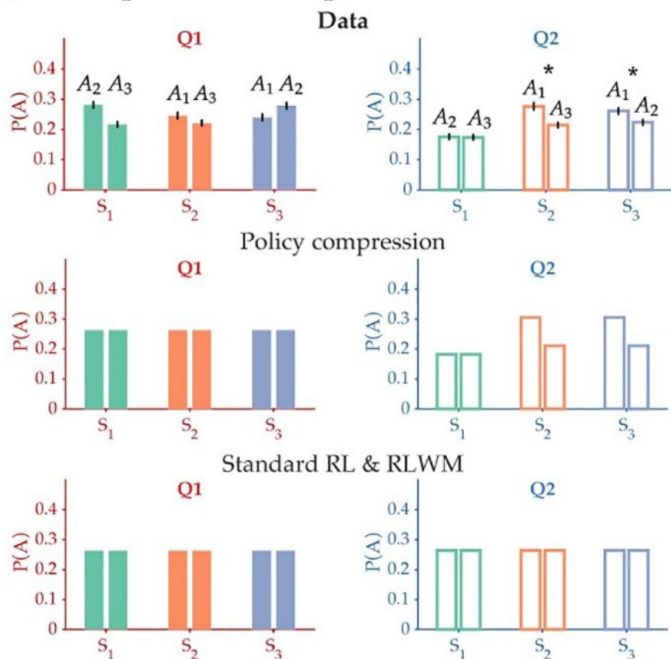
Task 1

B



C

Proportion of suboptimal actions chosen

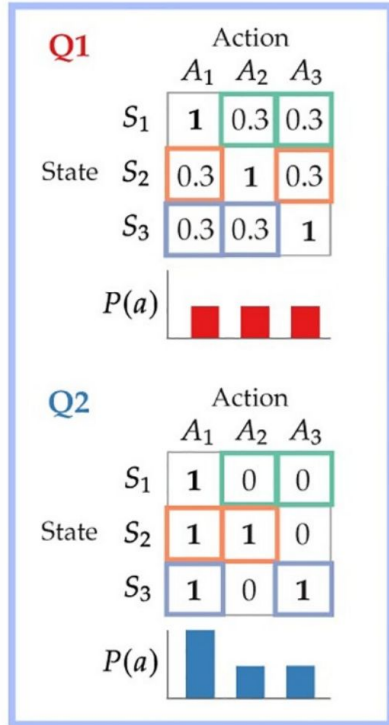


Q1 control, Q2 experimental (S^* appears 3X more often)

Does Action Frequency Bias Actions?

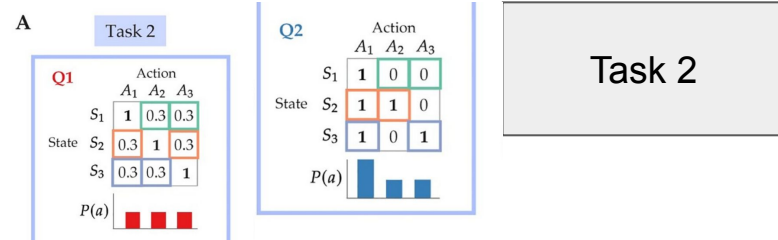
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Task 2

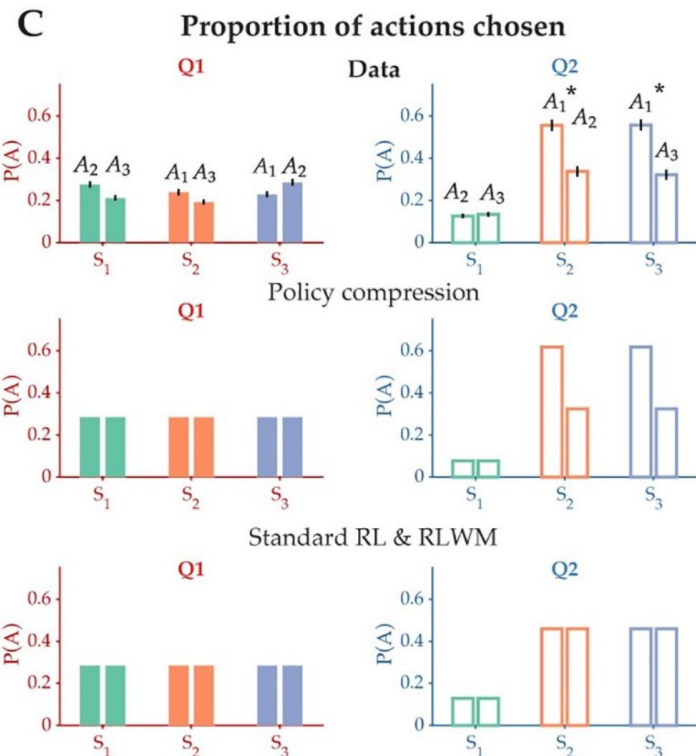
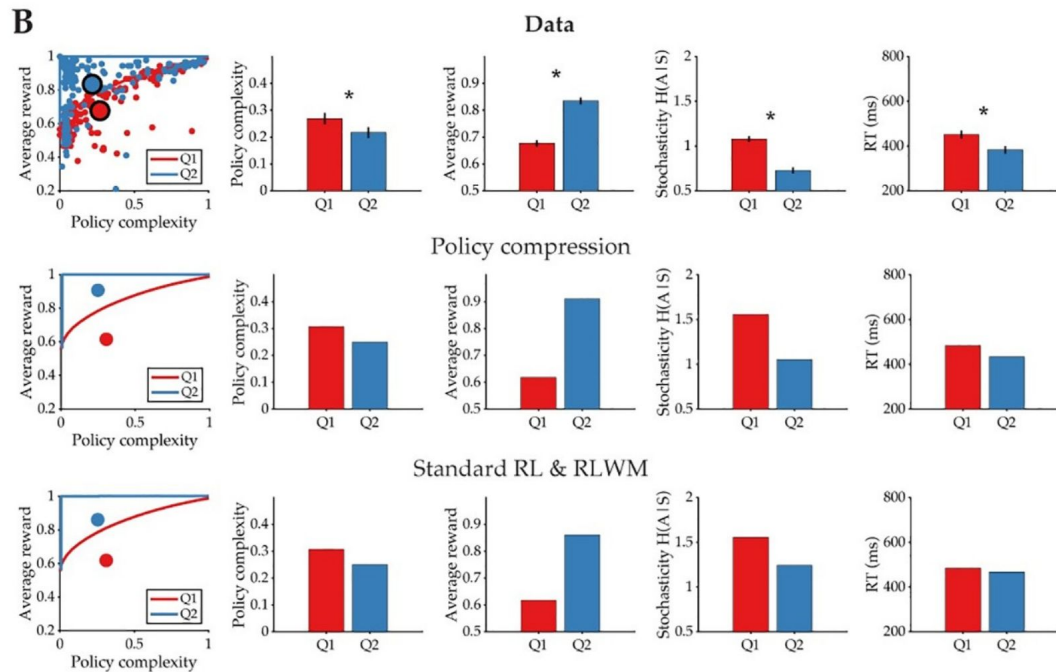


Q1 control,
Q2 experimental (S^* have **multiple** optimal responses)

Does Action Frequency Bias Action



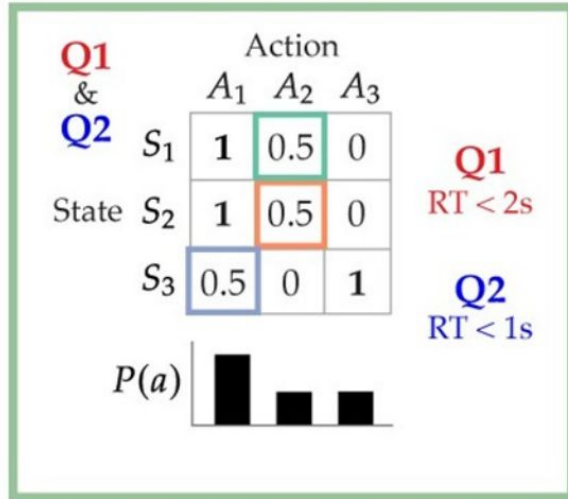
Task 2



Does Time Pressure Compress Policies?

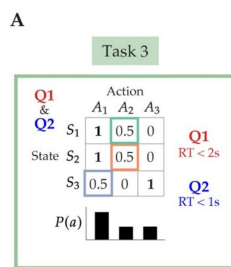
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Task 3

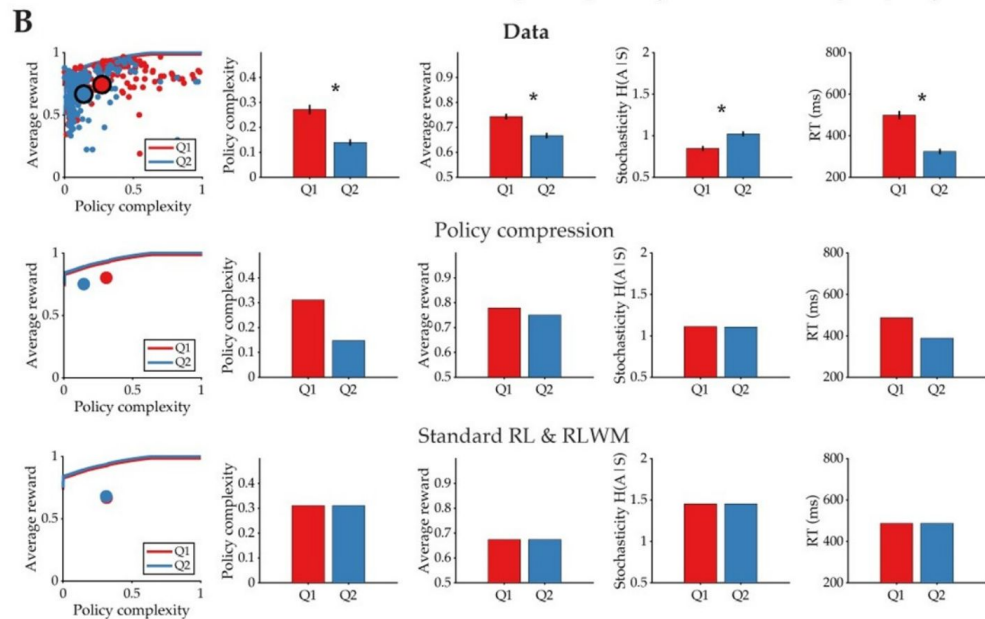


Q1 control (standard 2 second response period)
 Q2 experimental (1 second response period)

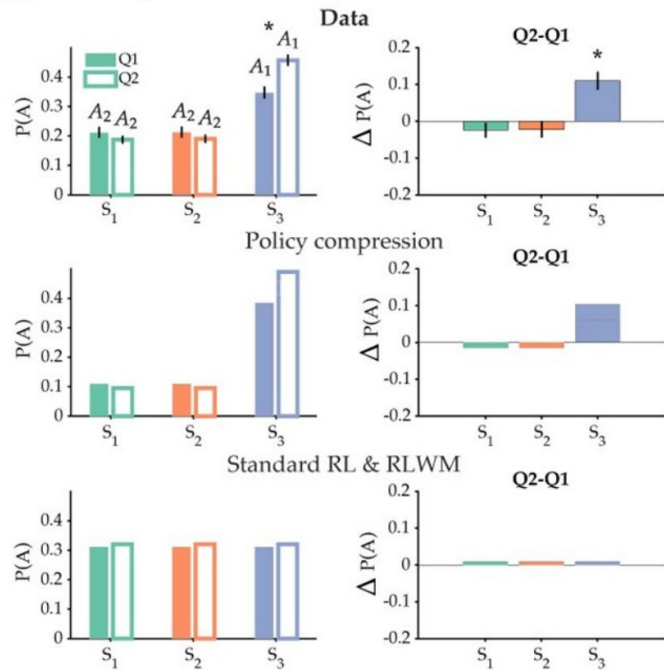
Does Time Pressure Compress Policies?



Task 3

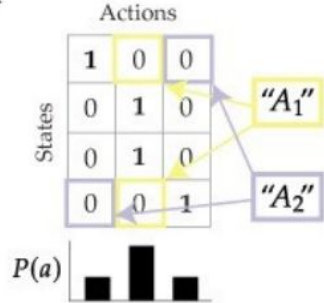


C Proportion of suboptimal actions chosen

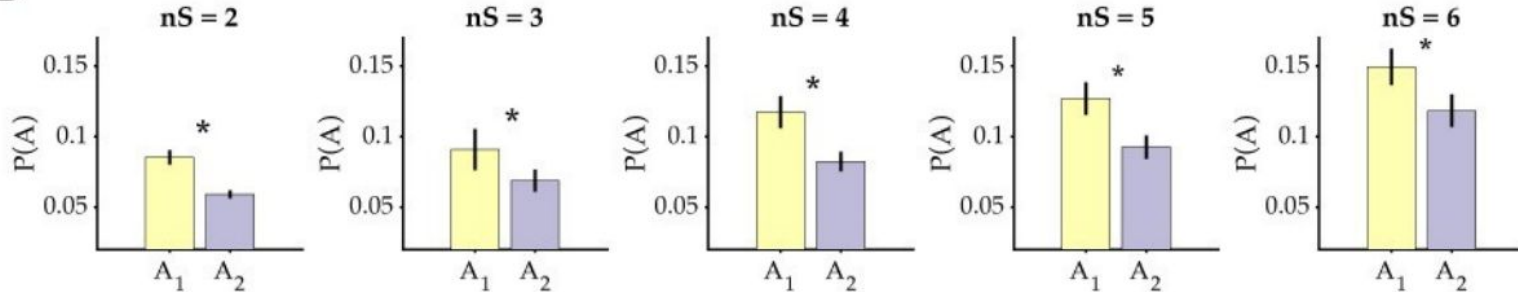


Does This Scale with Memory Load?

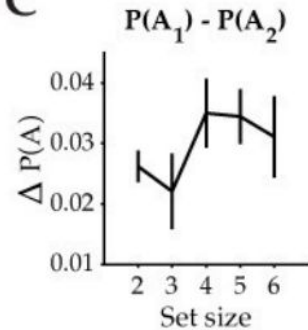
A Example from $nS = 4$



B

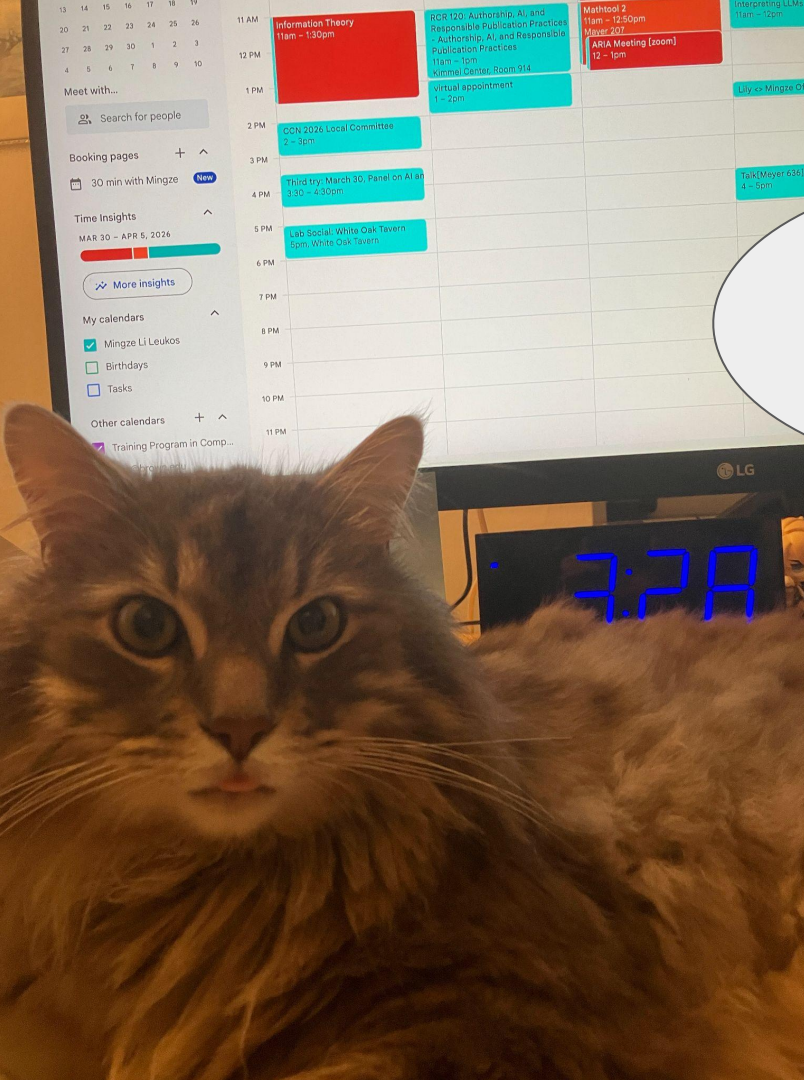


C



Novelties & Relevance

- **First** paper to **prospectively** design experiments to test policy compression
- compression is adaptive for low-capacity individuals, not just a source of error, connects to the broader resource-rationality program
- Enabled future work to design **decision environments** that work with people's compression biases.
- Policy Complexity measure = how much state information is used for action selection -> **attention??**



Thank You!